

Fixed-Length Paths I

Time limit: 1.00 s

Memory limit: 512 MB

Given a tree of n nodes, your task is to count the number of distinct paths that consist of exactly k edges.

Input

The first input line contains two integers n and k : the number of nodes and the path length. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: the number of paths.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 2 1 2 2 3 3 4 3 5	4